

Richwell Cyrille Santos Perez

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EDUCATION

University of Illinois Urbana-Champaign

Master of Computer Science

Bachelor of Science in Computer Science with Honors

August 2018 - May 2023

GPA: 3.52

GPA: 3.81

EXPERIENCE

AI Engineer, RAVE Aerospace

Laramie, WY; Brea, CA | June 2025 - Present

- Engineered a scalable RAG chatbot serving hundreds of internal users with Vue.js, Quart, and PostgreSQL on Azure App Services, incorporating managed identity authentication, async connection pooling, and structured logging for full observability
- Architected a hierarchical multi-agent LangGraph system with 16 custom LangChain tools and dual PostgreSQL databases, reducing query latency by 50% through optimized multi-agent routing, hybrid search, SSE streaming, and context engineering
- Created conversation analytics and evaluation pipelines with ML clustering and automated testing, leveraging a LangSmith harness with 6 evaluators across 40+ test cases to identify failure modes and improve AI agent performance
- Developed an LLM-powered flight log intelligence platform on MCP servers, enabling engineers and stakeholders to query logs via Claude and extract actionable insights
- Built large-scale data pipelines handling billions of records with Drain3 and PySpark, extracting ML features into Delta Lake for data analytics, predictive maintenance, and anomaly detection
- Designed interactive analytics to monitor device health checks, logs per seat, and flight-path/system data for IFE system troubleshooting

Database Administrator, Illinois Secretary of State

Springfield, IL | February 2025 - June 2025

- Managed high-throughput DB2 databases (5B+ records) on z/OS mainframes for statewide applications, optimizing complex SQL queries and schemas to reduce critical lookup times from ~20 seconds to near-instantaneous
- Implemented ETL pipelines and automation scripts to streamline data migration and reduce manual operational workload
- Collaborated with cross-functional teams to design efficient schemas, stored procedures, and Azure-based services for cloud-connected applications
- Supported development and deployment of statewide initiatives, including Apple mobile driver's licenses and REAL ID

Graduate Teaching Assistant, University of Illinois Urbana-Champaign

Urbana, IL | August 2022 - May 2023

- Led discussions and labs for CS222 Software Design & CS411 Database Systems, teaching 800+ students on design patterns and databases
- Collaborated with the professor and team to design, review, and grade projects, research papers, and assignments, ensuring academic integrity and consistency across multiple course sections
- Mentored students on full-stack web projects, providing guidance on tools, languages, and database integration to enhance practical application development

Software Engineer, University of Illinois Urbana-Champaign

Urbana, IL | July 2021 - August 2021

- Developed LabWindows/CVI software for new magnet mapping hardware and other physics lab applications, improving UI, fixing bugs, implementing control logic, and enabling real-time data interpretation

PROJECTS

UIUC Letter Grades | Full-Stack Web App with AI Integration

- Built a full-stack analytics platform with Node.js, MongoDB, and Python to visualize and analyze course data across 10+ years, enabling students to compare instructors and predict course outcomes
- Trained ML models (linear/logistic regression, sequential models) and visualizations for predictive grade analytics
- Implemented user authentication and automated data processing pipelines for efficient data handling

Video Anomaly Detection Model | Leveraging Weapon Recognition for Enhanced Accuracy

- Integrated weapon detection with anomaly scores, improving crime classification accuracy through multi-model decision-making
- Developed an AI pipeline using UCF-Crime anomaly detection & YOLO/Faster R-CNN for real-time crime recognition
- Optimized anomaly detection using Multi-Instance Learning (MIL) & improved loss functions for better predictions

Deep Learning Aerosol Model | 3rd Place, Ashby Prize in Computational Science, NCSA AI Innovation Hackathon

- Engineered a transformer and Encoder/Decoder models to capture temporal patterns in climate-relevant aerosol properties
- Applied advanced data preprocessing techniques to handle high-dimensional climate data and reduce the curse of dimensionality
- Leveraged HAL 9000 (UIUC's supercomputer) for high-performance training, optimizing deep learning workload scalability

SKILLS

Languages and Software: Python, Java, C/C++, JavaScript, SQL, MongoDB, PostgreSQL, Neo4j, Bash/Shell

AI/ML: TensorFlow, PyTorch, PySpark, Scikit-learn, LangChain, OpenCV, NLTK, Pandas, NumPy, Matplotlib

Frameworks and Tools: GCP, Azure, AWS, React.js, Node.js, Git, CI/CD, Docker, Postman, Microsoft Fabric, Delta Lake